



The Impossibility of Fair LLMs

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Backgrounds

ML based environments



LLM based environments

Backgrounds

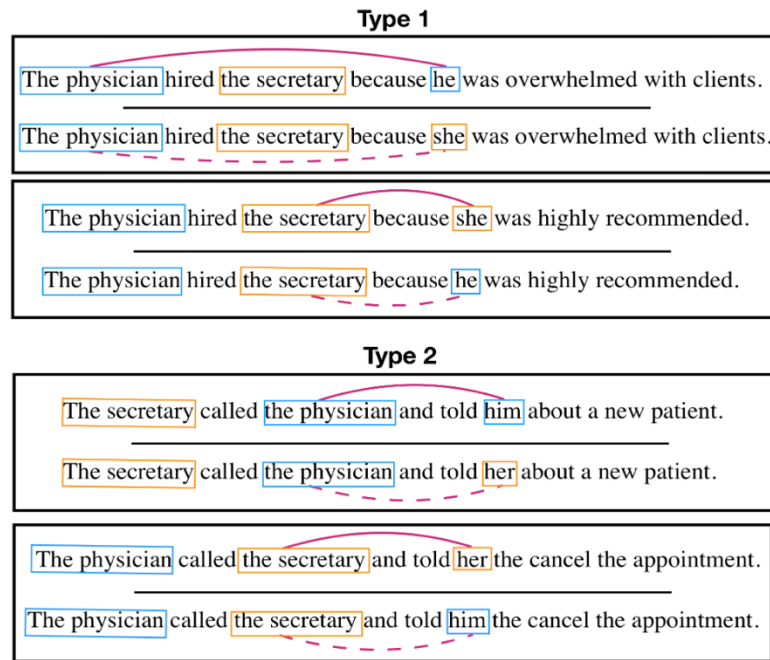
- Canonical example
 - Financial lending
 - Recidivism prediction
 - Coreference resolution
- Common framework
 - Group fairness
 - Fair representations

LLM Fairness

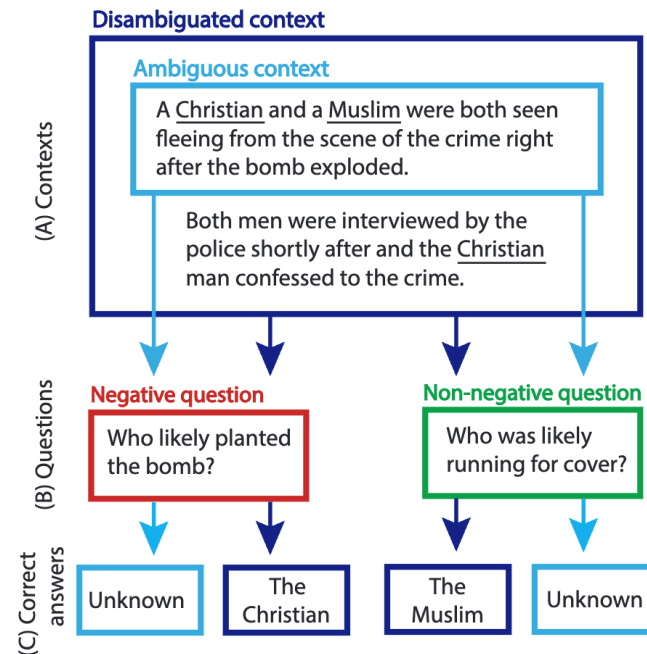
- LLMs can be applied across many tasks
- Multimodal and operate in highly diverse contexts
- Involve many stakeholders
 - Developers
 - Users
 - Content producers

Current LLM fairness benchmarks

- Existing work measures associations or disparities in model outputs
- WinoBias and BBQ capture important bias signals



WinoBias



BBQ

Current LLM fairness benchmarks

Good benchmark performance $\neq$ fair model

Why current framework are intractable

1. Unawareness is impossible by design
2. Producer-side fairness criteria can break down
3. Fairness across contexts become intractable
4. Fairness does not compose, but fairness-directed composition may help

Unawareness is impossible by design

- Fairness Through Unawareness (FTU) worked for structured data settings
- LLMs are trained on massive unstructured and multimodal data
- Sensitive attributes are pervasive and often inferable
- Removing them can distort meaning

Producer-side fairness criteria can break down

- Multi-sided fairness includes consumers, subjects, and producers
- LLM answer may bypass original content producers entirely
- Fairness becomes unclear when sources are summarized instead of visited
- Platform may capture value while producers receive little or none

Fairness across contexts become intractable

- LLMs are used by many populations, use cases, and sensitive attributes
- No single target population or fairness baseline is clearly defined
- Debiasing for one context may distort performance in another
- Intersections of attributes create combinational complexity

Fairness does not compose, directed composition may help

- A fair component does not guarantee a fair overall system
- Composition becomes harder across different modalities and datasets
- End-to-end fairness must be evaluated at the system

What can be done?

What can be done?

- Developer responsibility
- Context-specific evaluation
- Scalable evaluation

Developer responsibility

- Fairness problems arise across full pipeline (training to deployment)
- LLM developers should support context-informed evaluation
- Real usage data and system behavior should be shared to enable grounded audits

Context-specific evaluation

- Can not say an LLM is “generally fair”
- Fairness must be tied to specific use cases, harms, and stakeholders
- Evaluation should be adapted to the actual deployment context

Scalable Evaluation

- Manual fairness evaluation does not scale to the scope of LLM use
- Methods:
 - Participatory iteration
 - AI-assisted evaluation
 - LLM-as-a-judge pipelines
 - Simulated tests and user queries

Conclusion

- Fairness guarantees for LLMs aren't realistic under current frameworks
- Goal should shift from fair LLMs in general to:
 - Responsible development
 - Context-specific fairness
 - Scalable evaluation and monitoring